

YIYU LIU

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EDUCATION

Shanghai Jiao Tong University (SJTU), Shanghai, China

Aug 2021 – Jun 2025

Student in the **ACM Honor class**, GPA: **3.91/4.3**.

B.Eng. in Computer Science

Harvard University, Cambridge, MA, USA

Sep 2025 – Present

PhD Student in Computer Science, advised by **Prof. Juncheng Yang** and **Prof. Minlan Yu**.

PUBLICATIONS

FaaSMem: Improving Memory Efficiency of Serverless Computing with Memory Pool Architecture (ASPLOS'24)

Authors: Chuhao Xu, **Yiyu Liu**, Zijun Li, Quan Chen, Han Zhao, Deze Zeng, Qian Peng, Xueqi Wu, Haifeng Zhao, Senbo Fu, Minyi Guo.

TeleRAG: Efficient Retrieval-Augmented Generation Inference with Lookahead Retrieval

Co-first author. *In submission.*

RESEARCH EXPERIENCE

Research interests: cloud computing, operating systems.

Emerging Parallel Computing Center, SJTU, Shanghai, China

Sept 2023 – Jun 2024

Undergraduate researcher, advised by **Prof. Quan Chen**.

Topic: cloud computing

- (ASPLOS'24) Explored the memory footprint under serverless condition and proposed a new mechanism tailored for serverless containers to improve memory efficiency. The system yields up to 71.0% reductions in local memory usage and a maximum $2.2\times$ more deployment density.
 - **Role in the project:** designed the strategy to effectively and efficiently offload pages together with Chuhao; conducted experiments; wrote the paper with Chuhao.

**Paul G. Allen School of Computer Science & Engineering,
University of Washington, Seattle, WA, USA**

Jul 2024 – Dec 2024

Undergraduate researcher (student intern), advised by **Prof. Barış Kaşıkçı**.

Topic: cloud computing

- Built a disaggregated system to improve resource utilization while keeping the durability guarantee of persistent memory. A few key challenges were fixed in this project to improve the performance. This system outperforms NFS by up to $85\times$ on read-heavy workloads.
 - **Role in the project:** investigated the end-to-end procedure; made quite a few changes to the original system design; conducted experiments; wrote the paper with other authors.
- Optimized the performance of Retrieval-Augmented Generation from a system perspective. The latency of the system is optimized by up to $1.72\times$.
 - **Role in the project:** introduced a couple of improvements from the original project design; conducted core experiments; solved performance issue; wrote the paper with other authors.

**Harvard John A. Paulson School of Engineering and Applied Sciences,
Boston, MA, USA**

Sep 2025 – Present

PhD Student, advised by **Prof. Juncheng Yang** and **Prof. Minlan Yu**. *Topic: systems for machine learning*

- Working on designing systems to improve the efficiency of the prefix KV cache across requests for large language models.
 - **Role in the project:** Leading the project; designing the eviction strategy from the commonly used LRU and offline algorithms.

TEACHING EXPERIENCE

Programming	Aug 2022 – Jan 2023
Data Structure	Feb 2023 – Jun 2023
Principle and Practice of Computer Algorithms (project-based course)	Jun 2023 – Jul 2023
Compiler Design and Implementation (project-based course)	Aug 2023 – Jan 2024
Operating System	Feb 2024 – Jun 2024
Operating System	Feb 2025 – Jun 2025

Roll: teaching assistant. Work includes giving lectures, writing guidebooks or guiding documents, creating exam questions, etc.

SELECTED PROJECT

ACM Class Online Judgement System (🔗 [ACMClassOJ/TesutoHime](https://acm.sjtu.edu.cn/OnlineJudge)) Sep 2022 – Jun 2025

Website: <https://acm.sjtu.edu.cn/OnlineJudge>

Work: Wrote the development documentation for the whole project; added new features; fixed bugs; (Feb 2023 – Jun 2024) operated and maintained the online judge service (~369K submissions since 2020).

SKILLS

Programming languages: C, C++, Python, Go, Java, Kotlin, Verilog and Bash.

Markup languages: HTML, Markdown and $\text{\LaTeX}$.

LANGUAGES

English: fluent.

Chinese mandarin: native.